AKALANKA JAYAWARDHANE

IT Undergraduate | AI & Machine Learning | Computer Vision | Software Engineering

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PROFESSIONAL SUMMARY

IT undergraduate with hands-on experience in Artificial Intelligence, Machine Learning, Computer Vision, Software Engineering, and Web Development through academic and personal projects. Skilled in Python, TensorFlow, PyTorch, YOLO, OpenCV, React, JavaScript, and modern web technologies, with practical experience in developing machine learning models, image classification systems, computer vision applications, and software solutions. Possesses a strong problem-solving mindset and an ability to learn and adapt to new technologies. Interested in building intelligent, reliable, and user-focused applications that address practical real-world challenges.

CORE SKILLS

Programming Languages: Python, Java, JavaScript, C, C++
AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision, CNNs, Image Classification, Object Detection, YOLO
Web Technologies: HTML, CSS, JavaScript, React, Three.js, REST APIs
Mobile Development: Android Studio, Java, Kotlin
Database & Backend: MySQL, PostgreSQL, Supabase
Tools & Platforms: Git, GitHub, VS Code, Docker, Postman, Jupyter Notebook, Google Colab
Soft Skills: Problem Solving, Teamwork & Leadership, Communication & Presentation, Creativity, Adaptability
Languages: Sinhala, English

ACADEMIC PROJECTS & RESEARCH

2D Floor Plan to 3D Room & Furniture Arrangement System *2026*

- Developing an intelligent system that analyzes 2D floor plans and extracts walls, doors, windows, and room information
- Applying computer vision, object detection, image processing, and OCR techniques for floor-plan understanding.
- Developing a pipeline to transform detected floor-plan information into an interactive 3D room environment
- Implementing rule-based and intelligent furniture placement using room dimensions, spatial constraints, and user preferences.

Vehicle Classification System *2026*

- Developed a deep learning-based image classification system for identifying vehicle categories
- Prepared and preprocessed a custom dataset for model training and validation
- Built and trained a CNN model using TensorFlow/Keras.
- Worked on model evaluation and dataset optimization to improve classification performance.

FoodGo – Android Application *2026*

- Developed an Android-based food ordering application as a university project.
- Implemented application interfaces and core user interactions using Java and Android Studio.
- Applied software development and mobile application design principles in a practical project environment.

Music Player Application *2025*

- Developed a mobile music player application using Java.

EDUCATION

Bachelor of Science in Information Technology *2023 – Present*
Rajarata University of Sri Lanka

G.C.E. Advanced Level Examination *2011 – 2019*
Pinnawala Central Collage, Rambukkana